Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 3 — Trigonometric functions Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第3章 · 三角函数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C03-R01 · P3 · Chapter 3

An angle θ satisfies $\sin\theta = -5/13$ and $\pi < \theta < 3\pi/2$.

(a) Find the exact values of $\cos\theta$ and $\cot\theta$. (4)

(b) Find $\sec\theta + \operatorname{cosec}\theta$. (3)

(Total for Question 1 is 7 marks)

For $0 \leq x < 2\pi$, $\sec x = 2$.

(a) Find every exact solution. (3)

(b) Find every solution of $\sec^2 x = 4$ on the same interval. (3)

(Total for Question 2 is 6 marks)

Prove the identities on their common domains.

(a) $\sec^2 x - 1 = \tan^2 x.$ (3)

(b) $\dfrac{\sec^2 x - 1}{\sec^2 x} = \sin^2 x.$ (3)

(Total for Question 3 is 6 marks)

$\operatorname{cosec} x = -2$ for $-\pi \leq x \leq \pi$.

(a) Find all exact solutions. (3)

(b) Find the value of $\cot x$ at each solution. (3)

(Total for Question 4 is 6 marks)

5. P3-C03-R05 · P3 · Chapter 3

The curve is $y = \sec x$ for $-\pi \leq x \leq \pi$.

(a) Sketch it, showing all vertical asymptotes and points where $y = \pm 1$. (3)

(b) State the range and solve $\sec x \leq -2$ on the stated domain. (4)

(Total for Question 5 is 7 marks)

$\cot x = \sqrt{3}$ for $0 < x < 2\pi$.

(a) Find all exact solutions. (3)

(b) Find every excluded input of the function $\cot x$ on $[0, 2\pi]$. (2)

(Total for Question 6 is 5 marks)

$A = \arcsin(-\sqrt{3}/2)$ and $B = \arccos(-1/2)$.

(a) Find A, B exactly. (2)

(b) Find the exact value of $\cos A + \sin B$. (2)

(Total for Question 7 is 4 marks)

8. P3-C03-R08 · P3 · Chapter 3

$f(x) = \arcsin x$ for $-1 \leq x \leq 1$.

(a) Sketch its graph and state its range. (3)

(b) Find $\arcsin(\sin(5\pi/6))$ and explain why it is not $5\pi/6$. (3)

(Total for Question 8 is 6 marks)

9. P3-C03-R09 · P3 · Chapter 3

For $0 \leq x < 2\pi$, $2\sec^2 x - 5\sec x + 2 = 0$.

(a) Find all admissible values of $\sec x$. (3)

(b) Hence find all exact x. (3)

(Total for Question 9 is 6 marks)

On the common domain, $F(x) = \dfrac{\operatorname{cosec} x - \sin x}{\cos x}$.

(a) Show that $F(x) = \cot x$. (4)

(b) Solve $F(x) = 1$ for $0 < x < \pi$. (3)

(Total for Question 10 is 7 marks)

An angle $\theta = \arctan(5/12)$.

(a) Find $\sec\theta$ and $\operatorname{cosec}\theta$ exactly. (3)

(b) Find $\cos(\arctan(5/12))$ without a calculator. (2)

(Total for Question 11 is 5 marks)

$\operatorname{cosec}^2 x - 3\cot x - 3 = 0$ for $0 < x < \pi$.

(a) Write the equation as a quadratic in $\cot x$ and solve for this ratio. (3)

(b) Find the two solutions x in radians to 3 decimal places. (4)

(Total for Question 12 is 7 marks)

The graph is $y = \text{cosec}(2x)$ for $0 < x < \pi$.

(a) State all vertical asymptotes and the coordinates where $y = \pm 1$. (3)

(b) Solve $\text{cosec}(2x) = 2$. (3)

(Total for Question 13 is 6 marks)

For $-\pi/2 < x < \pi/2$, $\sec^2 x = 10$.

(a) Find the exact possible values of $\tan x$. (3)

(b) Given that $x < 0$, find x to 3 decimal places and $\sin x$ exactly. (4)

(Total for Question 14 is 7 marks)

Let $\alpha = \arccos(3/5)$.

(a) Find $\sin\alpha$ and $\cot\alpha$ exactly. (3)

(b) Find $\arccos(\cos(2\pi - \alpha))$ exactly in terms of α. (3)

(Total for Question 15 is 6 marks)

16. P3-C03-R16 · P3 · Chapter 3

For $0 < x < 2\pi$, $\cot^2 x = 1$.

(a) Find all exact solutions. (3)

(b) At which solutions is $\sec x$ negative? Give its exact value there. (3)

(Total for Question 16 is 6 marks)

On their common domain,

(a) prove $\dfrac{1+\cot^2 x}{\operatorname{cosec} x} = \operatorname{cosec} x$. (3)

(b) solve $\dfrac{1+\cot^2 x}{\operatorname{cosec} x} = -\sqrt{2}$ for $0 \leq x < 2\pi$. (3)

(Total for Question 17 is 6 marks)

The function $f(x) = \arccos x$ has domain $[-1, 1]$.

(a) State its range and sketch the graph, marking three exact points. (3)

(b) Solve $f(2x - 1) = \pi/3$, stating the input restriction. (3)

(Total for Question 18 is 6 marks)

For $0 \leq x < 2\pi$, $\sec x + \tan x = 3$.

(a) Use $(\sec x + \tan x)(\sec x - \tan x) = 1$ to find $\sec x, \tan x$. (3)

(b) Find the unique solution x in radians to 3 decimal places. (3)

(Total for Question 19 is 6 marks)

For $0 < x < \pi$, $\operatorname{cosec} x + \cot x = 2$.

(a) Find $\operatorname{cosec} x$ and $\cot x$ exactly. (3)

(b) Find x to 3 decimal places. (3)

(Total for Question 20 is 6 marks)

The curve is $y = \cot x$ for $0 < x < \pi$.

(a) Sketch the curve with asymptotes and intercept. (3)

(b) Find the exact interval on which $\cot x \geq 1$. (3)

(Total for Question 21 is 6 marks)

For $0 \leq x < 2\pi$, $\sec^2 x - 4\tan x = 5$.

(a) Find the possible values of $\tan x$. (3)

(b) State how many solutions occur in the interval and explain the count. (3)

(Total for Question 22 is 6 marks)

$\alpha = \arcsin(1/3)$.

(a) Find $\cos\alpha$ and $\tan\alpha$ exactly. (3)

(b) Find $\arcsin(\sin(\pi + \alpha))$ in terms of α. (3)

(Total for Question 23 is 6 marks)

For real x where defined, $F(x) = \sec^2 x + \operatorname{cosec}^2 x$.

(a) Show that $F(x) = \dfrac{1}{\sin^2 x \cos^2 x}$. (3)

(b) Find $F(\pi/6)$ exactly. (3)

(Total for Question 24 is 6 marks)

The graph is $y = 2\sec(x - \pi/3) + 1$.

(a) State its range. (3)

(b) Find every vertical asymptote in $0 \le x \le 2\pi$. (3)

(Total for Question 25 is 6 marks)

An angle θ lies in $(\pi/2, \pi)$ and satisfies $\tan\theta = -2$.

(a) Find the exact values of $\sec\theta$ and $\operatorname{cosec}\theta$. (3)

(b) Find $\arctan(\tan\theta)$ in terms of θ. (3)

(Total for Question 26 is 6 marks)

For $0 \leq x < 2\pi$, $\operatorname{cosec} x = \sec x$.

(a) Find all solutions. (3)

(b) Find the common reciprocal value at each solution. (3)

(Total for Question 27 is 6 marks)

$\beta = \arccos(-4/5)$.

(a) Find $\sin\beta$ and $\cot\beta$ exactly. (3)

(b) Find $\arccos(\cos(2\pi - \beta))$ and $\arcsin(\sin\beta)$ in terms of β. (3)

(Total for Question 28 is 6 marks)

For $0 < x < \pi$, $\text{cosec}^2 x + \cot x = 3$.

(a) Find all possible values of $\cot x$. (3)

(b) Find the exact solutions using inverse trigonometric notation where necessary. (3)

(Total for Question 29 is 6 marks)

The equation $\arcsin x + \arccos x = \pi/2$ is considered for $-1 \leq x \leq 1$.

(a) Using the principal ranges, explain why this identity holds. (4)

(b) Solve $2\arcsin x = \arccos x$ exactly. (3)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C03-R01 · 7 marks

(a) · 4 marks

$-12/13$ and $12/5$.

第三象限正弦余弦均负，平方和给cos=−12/13；cot=cos/sin。

M1	平方和与象限
A1	cos
M1	倒比
A1	cot

(b) · 3 marks

$-221/60$.

倒数分别−13/12与−13/5，通分相加−221/60。

M1	两个正确倒数
M1	通分
A1	精确值

Question 2

P3-C03-R02 · 6 marks

(a) · 3 marks

$x = \pi/3, 5\pi/3.$

等价cosx=1/2，两个合法角均不使cos0。

M1	倒数转余弦
M1	象限解
A1	两角

(b) · 3 marks

$\pi/3, 2\pi/3, 4\pi/3, 5\pi/3.$

平方方程允许sec=±2，新增cos=−1/2两角。

M1	正负分支
M1	完整象限
A1	四解

Question 3

P3-C03-R03 · 6 marks

(a) · 3 marks

$\sec^2 x - 1 = (1 - \cos^2 x)/\cos^2 x = \tan^2 x$.

统一为$\cos^2$分母，再用基本恒等式。

M1	统一分母
M1	平方恒等式
A1	目标式

(b) · 3 marks

$\tan^2 x / \sec^2 x = \sin^2 x$.

用(a)再以sin/cos和1/cos代入，cos因子抵消。

M1	使用(a)
M1	倒数定义化简
A1	结果

Question 4

P3-C03-R04 · 6 marks

(a) · 3 marks

$x = -5\pi/6, -\pi/6$.

sinx=−1/2，在负半圈内两点；sin0排除无影响。

M1	转正弦
M1	负角筛选
A1	两解

(b) · 3 marks

$\sqrt{3}$ at $-5\pi/6$; $-\sqrt{3}$ at $-\pi/6$.

分别余弦负、正，而正弦都负，cot符号随之相反。

M1	两余弦值与符号
M1	cot商
A1	配对结果

Question 5

P3-C03-R05 · 7 marks

(a) · 3 marks

Asymptotes $x = \pm\pi/2$; points $(-\pi, -1), (0, 1), (\pi, -1)$.

cos零点为竖直渐近线；各段曲线符号与cos相同，绝对值至少1。

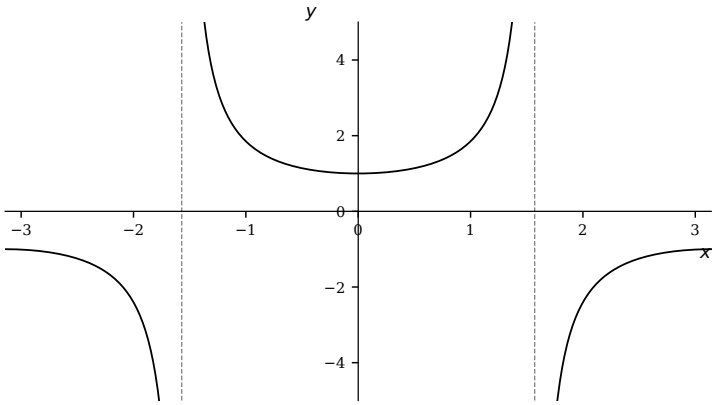
B1	渐近线
B1	三点
A1	正确三分支

(b) · 4 marks

Range $(-\infty, -1] \cup [1, \infty)$; $-2\pi/3 \le x < -\pi/2$ or $\pi/2 < x \le 2\pi/3$.

在cos负的分支，1/cos≤−2等价cos≥−1/2并保持负，按两段筛选，渐近线不取。

B1	值域
M1	负号不等式处理
M1	两分支端点
A1	完整范围



Suggested sketch

Question 6

P3-C03-R06 · 5 marks

(a) · 3 marks

$x = \pi/6, 7\pi/6.$

tanx=1/√3，周期π，两个同号象限。

M1	倒数比转换
M1	周期筛选
A1	两角

(b) · 2 marks

$0, \pi, 2\pi.$

cot=cos/sin，原分母sin零的点均无定义。

M1	原分母零点
A1	全部输入

Question 7

P3-C03-R07 · 4 marks

(a) · 2 marks

$A=-\pi/3,\ B=2\pi/3.$

arcsin主值域$[-\pi/2,\pi/2]$，arccos主值域$[0,\pi]$。

M1	正确两个主值范围
A1	两角

(b) · 2 marks

$(1+\sqrt{3})/2.$

$\cos(-\pi/3)=1/2$，$\sin(2\pi/3)=\sqrt{3}/2$。

M1	两精确三角值
A1	和

Question 8

P3-C03-R08 · 6 marks

(a) · 3 marks

Increasing from $(-1, -\pi/2)$ through $(0, 0)$ to $(1, \pi/2)$; range $[-\pi/2, \pi/2]$.

为正弦主分支关于y=x的反射，含两端点。

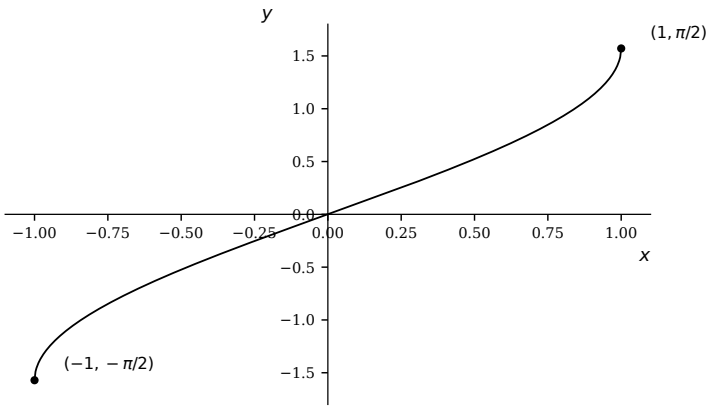
B1	三关键点
M1	递增反射形状
A1	值域

(b) · 3 marks

$\pi/6$; $5\pi/6$ lies outside the principal range of arcsine.

sin5π/6=1/2，arcsin必须取主值π/6。

M1	先算内部正弦
A1	主值
B1	主值限制理由



Suggested sketch

Question 9

P3-C03-R09 · 6 marks

(a) · 3 marks

$\sec x = 2$ only; reject $1/2$.

(2s−1)(s−2)=0，但实角正割绝对值至少1。

M1	二次分解
M1	正割值域筛选
A1	可行值

(b) · 3 marks

$\pi/3, 5\pi/3$.

cosx=1/2，取区间内两角。

M1	转余弦
M1	象限筛选
A1	两根

Question 10

P3-C03-R10 · 7 marks

(a) · 4 marks

$\operatorname{cosec} x - \sin x = (1 - \sin^2 x)/\sin x = \cos^2 x / \sin x$.

再除cosx得到cos/sin；原sin与cos都不能零。

M1	倒数与通分
M1	平方恒等式
M1	合法约分
A1	目标式

(b) · 3 marks

$x = \pi/4$.

cot=1给tan=1，主半圈仅π/4且不为原排除π/2。

M1	等价正切方程
M1	原域检查
A1	唯一角

Question 11

P3-C03-R11 · 5 marks

(a) · 3 marks

$13/12$ and $13/5$.

反正切给第一象限角，参考边5、12、13。

M1	参考三角形与主值
M1	倒数比
A1	两值

(b) · 2 marks

$12/13$.

由同一参考三角形读取余弦，不应把反三角视为倒数。

M1	正确函数嵌套解释
A1	精确值

Question 12

P3-C03-R12 · 7 marks

(a) · 3 marks

$\cot^2 x - 3\cot x - 2 = 0$; $\cot x = (3 \pm \sqrt{17})/2$.

用$\text{cosec}^2 = 1 + \cot^2$，整理并用二次公式。

M1	正确恒等替换
M1	二次公式
A1	两比值

(b) · 4 marks

$0.274, 2.082$ radians.

正cot在第一象限，负cot在第二象限；对负tan的反正切主值需加π。

M1	正确倒比
M1	负比分支加pi
M1	数值精度
A1	两角

Question 13

P3-C03-R13 · 6 marks

(a) · 3 marks

Asymptotes $x = 0, \pi/2, \pi$; points $(\pi/4, 1), (3\pi/4, -1)$.

sin2x零点给渐近线，±1处输入π/2、3π/2。

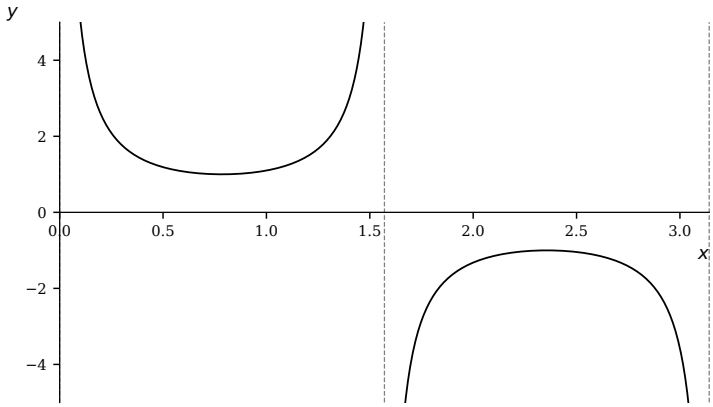
B1	三条渐近线
M1	极端输入缩放
A1	两点

(b) · 3 marks

$x = \pi/12, 5\pi/12$.

sin2x=1/2，输入在0至2π中两正半圈解，再除2。

M1	倒数转换
M1	输入范围
A1	两角



Suggested sketch

Question 14

P3-C03-R14 · 7 marks

(a) · 3 marks

$\tan x = \pm 3$.

$\tan^2=\sec^2-1=9$，两符号均在此区间可实现。

M1	平方恒等式
M1	两符号
A1	比值

(b) · 4 marks

$x = -\arctan 3 \approx -1.249$; $\sin x = -3/\sqrt{10}$.

此域cos正且x负，所以tan负；参考边−3与1，斜边$\sqrt{10}$。

M1	选负分支
A1	弧度值
M1	参考三角形符号
A1	正弦

Question 15

P3-C03-R15 · 6 marks

(a) · 3 marks

$4/5$ and $3/4$.

主值角在第一象限，勾股给sin4/5，cot比cos/sin。

M1	主值与平方和
A1	sin
A1	cot

(b) · 3 marks

α.

余弦在2π−α与α相等，而α本身位于arccos主值域。

M1	周期对称
M1	主值范围
A1	结果

Question 16

P3-C03-R16 · 6 marks

(a) · 3 marks

$\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.

cot=±1即tan=±1，四个参考π/4角。

M1	正负分支
M1	周期筛选
A1	四解

(b) · 3 marks

$3\pi/4, 5\pi/4$; $\sec x = -\sqrt{2}$.

第二、第三象限cos负；倒数$-\sqrt{2}$。

M1	正确象限
M1	精确倒数
A1	两角及值

Question 17

P3-C03-R17 · 6 marks

(a) · 3 marks

Numerator equals $\operatorname{cosec}^2 x$, and cosecant is never zero on its domain.

使用基本倒数恒等式，合法约去非零cosec。

M1	恒等替换
M1	约分合法
A1	目标式

(b) · 3 marks

$x = 5\pi/4, 7\pi/4$.

由(a)变为sinx=−1/√2，第三第四象限两个角；sin零点原本排除。

M1	引用(a)倒数转换
M1	象限与原域
A1	两解

Question 18

P3-C03-R18 · 6 marks

(a) · 3 marks

Range $[0,\pi]$; points $(-1,\pi),(0,\pi/2),(1,0)$; decreasing.

从余弦主分支反射得到反函数图，端点均包含。

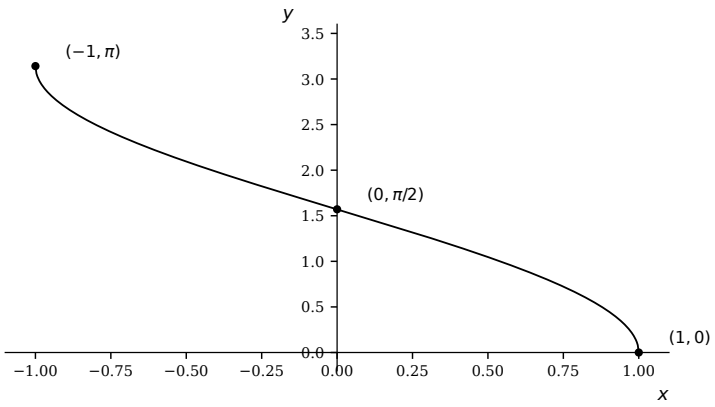
B1	三点
M1	递减形状
A1	值域

(b) · 3 marks

$0 \le x \le 1$; solution $x = 3/4$.

内部输入须−1⩽2x−1⩽1；2x−1=cosπ/3=1/2。

M1	复合定义域
M1	反解
A1	x



Suggested sketch

Question 19

P3-C03-R19 · 6 marks

(a) · 3 marks

$\sec x = 5/3$, $\tan x = 4/3$.

由恒等乘积另一因子$1/3$，两线性式相加减。

M1	找另一因子
M1	两式联立
A1	两比

(b) · 3 marks

$x = \arctan(4/3) \approx 0.927$.

sec正意味着cos正，tan正，唯一第一象限；不能加π。

M1	两比共同选象限
M1	反三角计算
A1	精度

Question 20

P3-C03-R20 · 6 marks

(a) · 3 marks

$5/4$ and $3/4$.

(cosec+cot)(cosec−cot)=1，另一因子1/2，相加减。

M1	共轭乘积
M1	联立
A1	两值

(b) · 3 marks

$x = \arctan(4/3) \approx 0.927$.

sin4/5、cot正，域内第一象限唯一。

M1	正确比值与象限
M1	计算
A1	弧度

Question 21

P3-C03-R21 · 6 marks

(a) · 3 marks

Asymptotes $x = 0, \pi$; intercept $(\pi/2, 0)$; strictly decreasing from $+\infty$ to $-\infty$.

cot=cos/sin，此半圈sin正而cos由正到负；标准图像每分支递减。

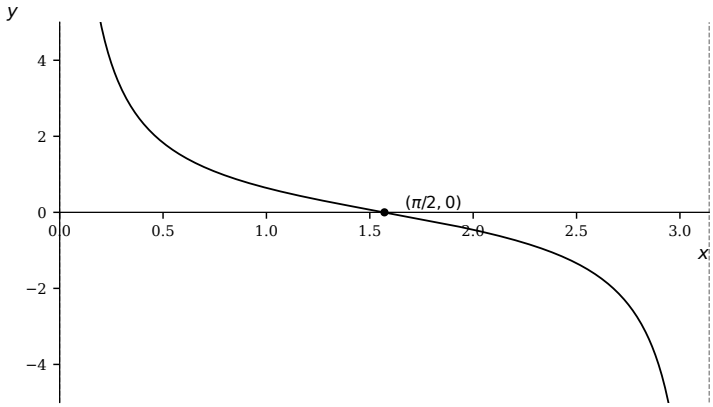
B1	渐近线
B1	零点
A1	正确分支

(b) · 3 marks

$0 < x \leq \pi/4$.

在本分支递减，等号π/4，左端为无定义渐近线。

M1	利用递减图像
M1	等号角和端点
A1	区间



Suggested sketch

Question 22

P3-C03-R22 · 6 marks

(a) · 3 marks

$\tan x = 2 \pm 2\sqrt{2}$.

以1+tan²替换sec²，得t²−4t−4=0。

M1	恒等替换
M1	二次公式
A1	两比

(b) · 3 marks

Four; each real tangent value occurs twice in a full interval of length 2π.

两不同比值，正切周期π，每一值在两个分支各出现一次，均不在渐近线。

M1	两个不同比值
M1	正切周期计数
A1	总数

Question 23

P3-C03-R23 · 6 marks

(a) · 3 marks

$2\sqrt{2}/3$ and $\sqrt{2}/4$.

主值角锐，cos=√(1−1/9)，tan为(1/3)/(2√2/3)，有理化。

M1	主值正根
M1	比值与有理化
A1	两值

(b) · 3 marks

$-\alpha$.

正弦(π+α)=−sinα，而−α位于arcsin主值区间。

M1	半周对称
M1	主值选择
A1	结果

Question 24

P3-C03-R24 · 6 marks

(a) · 3 marks

Common numerator $\sin^2 x + \cos^2 x = 1$.

以$\sin^2\cos^2$通分，原sin和cos均非零。

M1	正确公分母
M1	平方和
A1	目标式

(b) · 3 marks

$16/3$.

$\sin^2$=1/4、$\cos^2$=3/4，积3/16，取倒数。

M1	精确平方比
M1	倒数
A1	值

Question 25

P3-C03-R25 · 6 marks

(a) · 3 marks

$(-\infty,-1]\cup[3,\infty)$.

sec值域两支≤−1或≥1，乘正2加1得到≤−1或≥3。

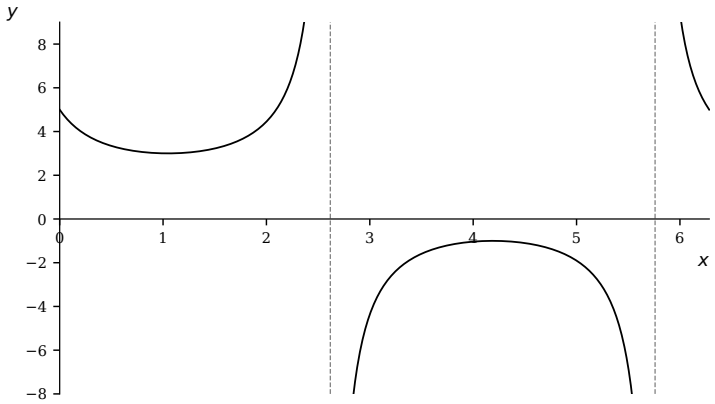
M1	原正割两支范围
M1	纵变换
A1	新范围

(b) · 3 marks

$x=5\pi/6,11\pi/6$.

cos(x−π/3)=0给x−π/3=π/2+kπ，筛选两个。

M1	分母零点
M1	输入相移与周期
A1	两线



Suggested sketch

Question 26

P3-C03-R26 · 6 marks

(a) · 3 marks

$-\sqrt{5}$ and $\sqrt{5}/2$.

参考两边2、1，斜边$\sqrt{5}$；第二象限cos负、sin正，取对应倒数。

M1	参考三角形
M1	象限符号
A1	两值

(b) · 3 marks

$\theta - \pi$.

反正切主值在$(-\pi/2, \pi/2)$，将第二象限θ减π进入负主值分支。

M1	正切周期
M1	主值区间
A1	表达式

Question 27

P3-C03-R27 · 6 marks

(a) · 3 marks

$x = \pi/4, 5\pi/4$.

原sin、cos都非零，可交叉相乘得sin=cos，tan1，两个周期解。

M1	约束下合法清分母
M1	正切与周期
A1	两角

(b) · 3 marks

$\sqrt{2}$ at $\pi/4$; $-\sqrt{2}$ at $5\pi/4$.

第一与第三象限的sin和cos分别$\pm\sqrt{2}/2$，倒数对应$\pm\sqrt{2}$。

M1	象限精确比
M1	倒数
A1	配对值

Question 28

P3-C03-R28 · 6 marks

(a) · 3 marks

$3/5$ and $-4/3$.

arccos主值第二象限，正弦为正3/5，cot负4/3。

M1	主值与平方和
A1	sin
A1	cot

(b) · 3 marks

β and $\pi - \beta$.

余弦补周返回原主值β；正弦主值必须是第一象限补角π−β。

M1	余弦主值
M1	正弦主值折回
A1	两结果

Question 29

P3-C03-R29 · 6 marks

(a) · 3 marks

$1, -2$.

$1+t^2+t=3$给$(t+2)(t-1)=0$。

M1	恒等替换
M1	因式分解
A1	两值

(b) · 3 marks

$x = \pi/4$ and $x = \pi - \arctan(1/2)$.

$\cot 1$给$\pi/4$；$\cot -2$对应$\tan -1/2$，在半圈内第二象限$\pi-\arctan 1/2$。

M1	倒比转换
M1	合法分支
A1	两精确表达式

Question 30

P3-C03-R30 · 7 marks

(a) · 4 marks

If $\alpha = \arcsin x \in [-\pi/2, \pi/2]$, then $\pi/2 - \alpha \in [0, \pi]$ and its cosine is x.

互余关系与两个主值范围共同唯一确定arccosx；不能只凭函数名消去。

M1	定义反正弦主值
M1	互余余弦关系
M1	主值唯一性
A1	恒等式

(b) · 3 marks

$x = 1/2$.

令α=arcsinx，用(a)得2α=π/2−α，α=π/6，回求sinα。

M1	使用恒等式线性求角
M1	取合法主值
A1	x